Final Project Report
COMS 4995: Causal Inference 2

Amy Nguyen Pham
CUID: nn2571
Mentors: Kevin Xia and Kasra Jalaldoust

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Project: Causal Transportability of Harmful Content Classifications across Different Social Media Domains

Topic: Generalizability, Robustness, and Adaptation

Abstract

Social media platforms are central to public discourse but are also vulnerable to harmful content such as hate speech. Although automated classifiers are widely used to detect such content, they are typically trained on a single platform and often overfit to spurious domain-specific correlations. Using causal graphs, we show that these associations are not transportable across domains, leading to poor generalization. In contrast, the causal effect, which is unconfounded by platform specific biases, can be transported under certain assumptions and offers a more robust signal for classification **(Section 3)**. This insight motivates our algorithm to estimate the causal effect for hate speech detection, enabling generalization across platforms **(Section 4)**. We then run our algorithm on a hate speech dataset and observe a modest improvement in accuracy using the causal classifier compared to the traditional approach **(Section 5)**.

Please refer to `https://github.com/amynguyenpham/CI2_Project/tree/main` for the data as well as the code used in the project experiment.

1 Introduction

Social media platforms have become critical venues for public discourse, but they are also prone to the proliferation of harmful content, such as hate speech, extremist rhetoric, and misinformation. Automated classifiers are increasingly deployed to identify and mitigate such content; however, these classifiers are often trained on data from a single platform and tend to rely on domain-specific, spurious correlations that reflect bias. For example, Perspective, Google's API to detect hate speech was trained on Wikimedia data and reported high accuracy in training. However, when the model was released, people noticed examples of bad predictions. For example, the single-term comment 'Arabs' was classed as 63% toxic, while the The phrase 'I love f****' (a horrible term) was only 3% toxic [1]. Similarly, a classifier trained on Twitter data—which features short, informal posts with hashtags and abbreviations—may overfit to spurious, succinct signals. When applied to Reddit, where posts are longer and more context-rich, the model may fail to generalize.

Hate speech classification models often struggle to generalize effectively [2] not only because of limited diversity in training data (for example, dialects and slang specific to a platform) or covariate shift but also due to the linguistic diversity through which hate speech is expressed across different social media platforms. Using Reddit and Twitter as an example, here is how offensive text (an indication that a protected class should not have voting rights) about the same topic manifests in the two platforms:

1. Twitter Example: Note that the platform's text limit and policy affect the linguistic style of hate speech. Tweets are often short, informal, and full of abbreviations, slang, hashtags, and emojis, as well as the use of uppercase/ lowercase, as Twitter has a 280-word limit and tweets are optimized for viral factors.

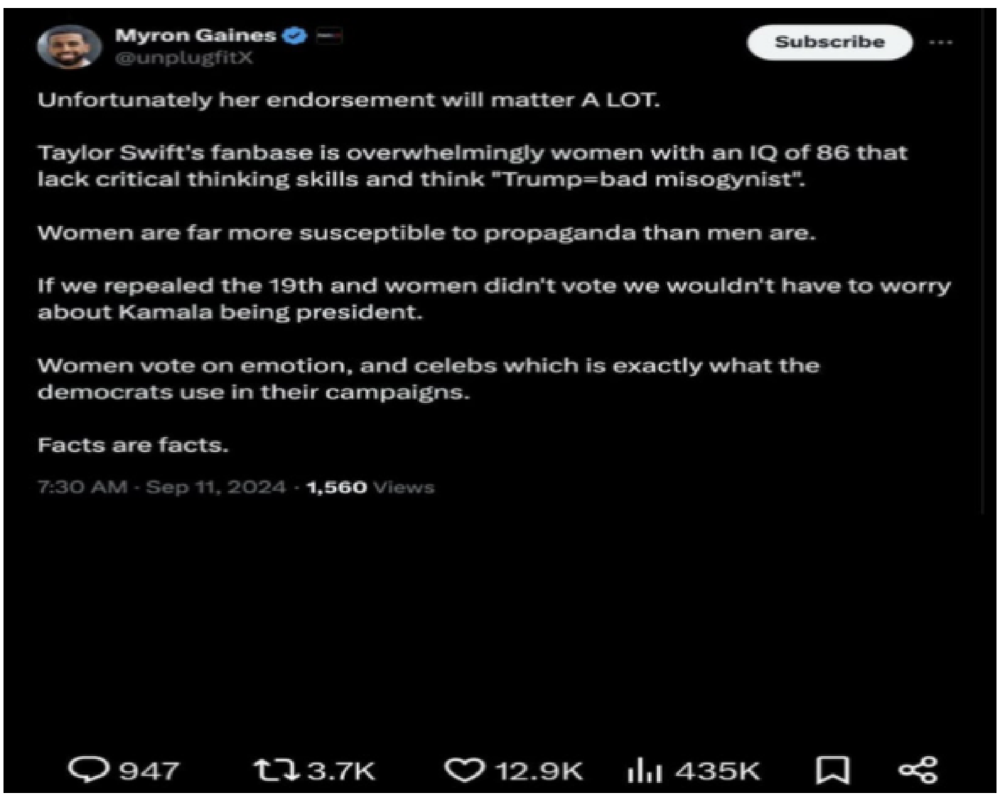
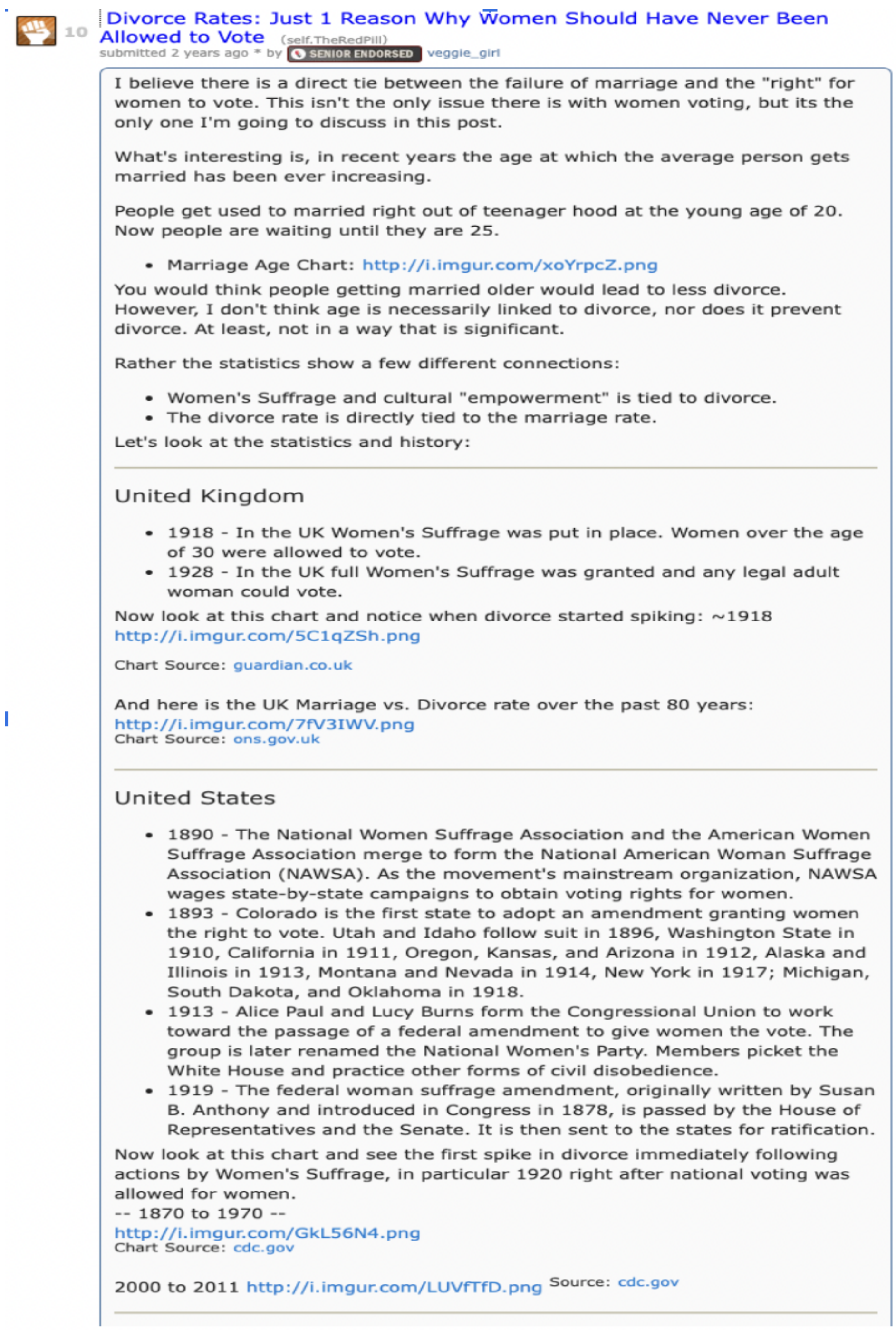


Figure 1: Descriptive caption about the image.

2. Reddit Example: Unlike Twitter, Reddit hateful posts are often longer, more structured, with either informal or academic style and often masked in pseudo-intellectual or sarcastic tones.


10 Divorce Rates: Just 1 Reason Why Women Should Have Never Been Allowed to Vote (self.TheRedPill)
submitted 2 years ago * by SENIOR ENDORSED veggie_girl

I believe there is a direct tie between the failure of marriage and the "right" for women to vote. This isn't the only issue there is with women voting, but its the only one I'm going to discuss in this post.

What's interesting is, in recent years the age at which the average person gets married has been ever increasing.

People get used to married right out of teenager hood at the young age of 20. Now people are waiting until they are 25.

- Marriage Age Chart: http://i.imgur.com/xoYrpcZ.png

You would think people getting married older would lead to less divorce. However, I don't think age is necessarily linked to divorce, nor does it prevent divorce. At least, not in a way that is significant.

Rather the statistics show a few different connections:

- Women's Suffrage and cultural "empowerment" is tied to divorce.
- The divorce rate is directly tied to the marriage rate.

Let's look at the statistics and history:

United Kingdom

- 1918 - In the UK Women's Suffrage was put in place. Women over the age of 30 were allowed to vote.
- 1928 - In the UK full Women's Suffrage was granted and any legal adult woman could vote.

Now look at this chart and notice when divorce started spiking: ~1918
http://i.imgur.com/5C1qZSh.png

Chart Source: guardian.co.uk

And here is the UK Marriage vs. Divorce rate over the past 80 years:
http://i.imgur.com/7fV3IWV.png
Chart Source: ons.gov.uk

United States

- 1890 - The National Women Suffrage Association and the American Women Suffrage Association merge to form the National American Woman Suffrage Association (NAWSA). As the movement's mainstream organization, NAWSA wages state-by-state campaigns to obtain voting rights for women.
- 1893 - Colorado is the first state to adopt an amendment granting women the right to vote. Utah and Idaho follow suit in 1896, Washington State in 1910, California in 1911, Oregon, Kansas, and Arizona in 1912, Alaska and Illinois in 1913, Montana and Nevada in 1914, New York in 1917; Michigan, South Dakota, and Oklahoma in 1918.
- 1913 - Alice Paul and Lucy Burns form the Congressional Union to work toward the passage of a federal amendment to give women the vote. The group is later renamed the National Women's Party. Members picket the White House and practice other forms of civil disobedience.
- 1919 - The federal woman suffrage amendment, originally written by Susan B. Anthony and introduced in Congress in 1878, is passed by the House of Representatives and the Senate. It is then sent to the states for ratification.

Now look at this chart and see the first spike in divorce immediately following actions by Women's Suffrage, in particular 1920 right after national voting was allowed for women.
-- 1870 to 1970 --
http://i.imgur.com/GkL56N4.png
Chart Source: cdc.gov

2000 to 2011 http://i.imgur.com/LUVfTfD.png Source: cdc.gov

Figure 2: Descriptive caption about the image.

The above example highlights how state-of-the-art hate speech detection models—such as transformer-based architectures and contrastive learning—often fail to generalize across platforms. These approaches tend to overfit to platform-specific cues rather than capturing the underlying causal relationship between text content and hate speech labels. Hence, in this project, I want to explore text classification through a causal lens. Prior research has demonstrated the effectiveness of causal transportability approaches for out-of-distribution classification in computer vision [3]. Given that language is continuously evolving and new social media domains are emerging, it is crucial to investigate how harmful content classifiers can be adapted to new environments. This adaptation is particularly important considering the substantial financial and emotional costs associated with training and experimenting on harmful content [4].

2 Related Work

2.1 Hate Speech Classification Task

Hate speech classification faces significant challenges due to linguistic ambiguity, evolving platform specific norms, and distributional shifts across social media sites. A challenging goal of this task is to build models that not only perform well in domain but also generalize across environments. Different approaches have been proposed to address this challenge. For example, some techniques such as HateBERT [5] use a large language model such as BERT and fine-tune it with hateful data from specific platforms. While large language models benefit from training on vast datasets and offer strong generalization capabilities, they remain susceptible to overfitting the pre-training corpus and inheriting platform-specific biases embedded in the training data.

2.2 Causal Inference and Transportability Theory

Causal inference enables reasoning about generalization across domains [6]. We begin by modeling the generative process between hate speech text and its label using a *Structural Causal Model* (SCM) [6, Ch. 7]. Each SCM $\mathcal{M}$ is defined as a 4-tuple $\langle \mathbf{V}, \mathbf{U}, \mathbf{F}, P(\mathbf{U}) \rangle$, where:

- $\mathbf{V}$ is the set of observed variables (e.g., text X and label Y),
- $\mathbf{U}$ represents the unobserved (exogenous) variables,
- $\mathcal{F}$ is the set of deterministic structural assignments, such that each $V_i \in \mathbf{V}$ is generated by $V_i \leftarrow f_i(\text{pa}_i, U_i)$,
- $P(\mathbf{U})$ is the joint distribution over the unobserved variables.

To capture cross-domain invariance in the hate speech generation process, we adopt the framework of selection diagrams and the theory of causal transportability, which formally characterize how and when causal relationships can be validly transferred between domains [7,8].

3 Problem Formulation: Hate Speech Detection Through a Causal Lens

3.1 Structural Modeling of Classification Task

Let X denote the input text and $Y \in \{0,1\}$ the binary hate speech label, with specific instances x and y. The goal is to predict $Y = y$ given $X = x$. Traditionally, in a probabilistic framework, this corresponds to estimating $P(Y \mid X)$ from training data and selecting the label at inference time via $\boldsymbol{\arg\max_y P(Y = y \mid X = x)}$.

Standard classifiers learn $P(Y \mid X)$ from a source platform π and are deployed on a different platform π^*, where performance often degrades due to platform-specific factors such as distributional shifts in topic, style, or sentiment [9,10,11]. We denote such platform-specific factors as A, which influence both X and Y, encompassing latent attributes like topic, formality, and sentiment.

We adopt a causal framework and represent the data-generating process using a Structural Causal Model (SCM) as discussed in **Section 2**. Here, the observed variables include X, Y, and A, while U denotes latent exogenous variables. The system is defined by structural functions $\mathcal{F} = \{f_A, f_X, f_Y\}$, such that:

$$A \leftarrow f_A(U_A), \quad X \leftarrow f_X(U_X, A), \quad Y \leftarrow f_Y(X, A, U_Y)$$

with $P(U)$ denoting the joint distribution over U.

In traditional classification, train and test data are assumed to come from the same domain. However, in cross-platform settings, this assumption fails. This setting is formalized as the *transportability problem* in causal inference [7,8], where training data come from domain π and test data from π^*. We assume that the labeling mechanism f_Y and text generation f_X, as well as distribution $P(U_X)$ and $P(U_Y)$ are invariant across domains, while the platform-specific mechanisms f_A and $P(U_A)$ may vary.

To capture domain shifts, we introduce a **selection variable** S that encodes changes across domains. For instance, if A denotes the topic, Twitter (π) may contain more immigration or political content, while Reddit (π^*) features a broader mix including tourism. As a result, $P(A = \text{“Immigration”}) \neq P^*(A = \text{“Immigration”})$, and we model this shift by directing $S \rightarrow A$. Due to node S, A, even when X is identical, differences in A across domains can lead to non-transportable associations between X and Y.

Putting it together, we come up with the causal graph as below:

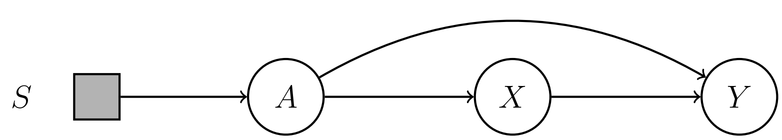


Figure 3: Causal graph for hate speech generation

When both training and deployment occur on the same platform, learning $P(Y \mid X)$ is a sensible approach, as it leverages the full available information to optimize prediction accuracy. However, this approach breaks down in cross-domain settings. Because the model is trained exclusively on data from the source domain π, it captures $P(Y \mid X)$ specific to that environment, which may diverge from the true distribution $P^*(Y \mid X)$ in the target domain π^*.

Proposition 1: Let $\mathcal{M}$ and $\mathcal{M}^*$ be the Structural Causal Models (SCMs) representing the source and target domains π and π^*, respectively, and assume they are compatible with the causal graph shown in **Figure 3**. Then,

$$P^*(Y \mid X) \neq P(Y \mid X).$$

In words, the classifier defined by $P(Y \mid X)$, learned in domain π, is *not transportable* to the target domain π^*, and cannot be used to make valid inferences about $P^*(Y \mid X)$. This discrepancy arises due to unobserved confounding between X and Y via A, which introduces spurious correlations that differ across domains.

Proposition 2: Let $\mathcal{M}$ and $\mathcal{M}^*$ be the Structural Causal Models (SCMs) representing the source and target domains π and π^*, respectively, and assume they are compatible with the causal graph in **Figure 3**. Then:

$$P^*(Y \mid \text{do}(X)) = \sum_a P(Y \mid X, A) \cdot P^*(A)$$

Proof:

$$P^*(Y \mid \text{do}(X)) = \sum_a P^*(Y \mid X, a) \cdot P^*(a) \quad \text{(Back-Door Adjustment)}$$

Since Y is d-separated from the selection node S given X and A, the conditional distribution $P(Y \mid X, A)$ is transportable across domains as conditioning on X,A prevents any change in S to affect Y. Hence,

$$P(Y \mid X, A) = P^*(Y \mid X, A).$$

Therefore,

$$P^*(Y \mid \text{do}(X)) = \sum_a P(Y \mid X, a) \cdot P^*(a).$$

For concreteness, consider the following example that illustrates how $P(Y \mid X)$ is not transportable across domains, while $P^*(Y \mid \text{do}(X))$ may serve as a more reliable basis for classification in the target domain:

Example 1. Let π denote the source domain and π^* the target domain. Suppose we are given the unobserved true distributions $P(v)$ and $P^*(v)$ as follows:

X	**a**	**Hate_Label**	**a***	**Hate_Label***
"People need to leave"	0	1	0	1
"People need to leave"	1	0	1	0
"People need to leave"	0	0	0	0
"People need to leave"	1	1	1	1
"People need to leave"	1	1	0	0
"People need to leave"	1	1	0	0
"People need to leave"	1	1	0	0
"People need to leave"	1	1	0	0
"Travel is fun"	X	X	0	0
"Travel is fun"	X	X	0	0
"Travel is fun"	X	X	0	0
"Travel is fun"	X	X	0	0

Table 1: Empirical distributions $P(v)$ and $P^*(v)$

Assume $A \in \{0, 1\}$, where 0 represents *others* and 1 represents *immigration*, and $Y \in \{0, 1\}$, where 0 denotes a *non-hate label* and 1 denotes a *hate label*. An "X" symbol indicates that the text x does not appear in a given domain (e.g., "Travel is fun" does not exist in π). Probabilistically, texts labeled as *immigration* are more likely to be associated with hate speech than those labeled as *others*.

Let us refer to "People need to leave" as x_0. From the data, we note that:

$$P(Y = 1 \mid X = x_0) = \frac{3}{4} = 0.75$$

$$P^*(Y = 1 \mid X = x_0) = \frac{1}{4} = 0.25$$

In this case, a classifier trained on $P(Y \mid X)$ will not perform well when deployed in the π^* environment.

However, we also observe:

$$P^*(Y = 1 \mid \text{do}(X = x_0)) = \sum_a P^*(Y \mid X = x_0, a) \cdot P^*(a)$$

$$= \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{5}{6} = 0.22$$

Hence, in this example, a classifier trained to estimate $P^*(Y \mid \text{do}(X))$ will be a better approximate classifier of the target environment than one trained on $P(Y \mid X)$. We will then explore using $P^*(Y \mid \text{do}(X))$ as the basis for our hate speech classifier in the rest of the project.

3.2 Symbolic Transportability Task

Given that $P^*(Y \mid \text{do}(X))$ will be a better approximate classifier of the target environment, we now formally define the causality task to identify $P^*(Y \mid \text{do}(X))$ as follows:

$$\mathscr{I} = \langle P^*(y \mid \text{do}(x)),\ \mathbb{P},\ \mathscr{D}^{\triangle} \rangle,$$

where $\mathbb{P} = \{P(\mathbf{V})\}$, and $\mathscr{D}^{\triangle}$ is described in **Figure 3**.

Per **Proposition 2**:

$$P^*(Y \mid \text{do}(x)) = \sum_a \underbrace{P(Y \mid x, a)}_{\text{from } \mathbb{P} \text{ (source domain)}} \cdot \underbrace{P^*(a)}_{\text{from } \mathbb{P}^* \text{ (target domain)}}$$

We can readily obtain $P^*(Y \mid x, a)$ from the training data. In the next section, we demonstrate how to calculate the above quantity, as well as estimate $P^*(A)$ from $P(\mathbf{V})$ under certain causal assumptions.

4 Deriving Causal Estimand for Target Environment

Building on the previous observation that $P(Y \mid \text{do}(X))$ serves as a suitable proxy for the classifier in the target domain, i.e., $P^*(Y \mid X)$, we now discuss the assumptions required to estimate this quantity.

4.1 Constructing P*(A)

Assumption 1 (Invariant): The process of generating the label A from text is assumed to be invariant across platforms, such that $P(A \mid X) = P^*(A \mid X)$.

This assumes that given a text X, the conditional distribution over labels A (e.g., topic, style, or sentiment) remains stable across domains. The rationale is that A reflects intrinsic properties of the text, such as its semantic content or linguistic tone, which are primarily determined by X itself rather than by external platform specific factors. For instance, the input "People need to leave" plausibly maps to a limited set of topics such as Immigration or Tourism, regardless of whether it appears on Twitter or Reddit. Although not universally true, this assumption holds approximately in many practical classification tasks.

Under this assumption, we train a stochastic generator to sample label a for each input x multiple times, and apply Monte Carlo sampling to estimate the target distribution $P^*(A)$.

4.1.1 Building Generator G(x) = a

To construct the stochastic generator $G(a \mid x)$, we leverage large language models (LLMs) using a one-shot prompting strategy. Specifically, we use the `google/flan-t5-base` model from Hugging Face to generate topic labels A given an input text X [12]. The prompt includes a list of candidate labels (e.g.,race, religion, immigration, others) to guide the model. At inference time, for each text x, we generate multiple topic samples using nucleus sampling and temperature-controlled decoding to approximate the conditional distribution $P(A \mid X)$. This process is inherently stochastic: the `model.generate` function is configured with `do_sample=True`, along with sampling parameters such as `top_k`, `top_p`, and `temperature`, which introduce controlled randomness into the decoding process. As a result, each invocation of the generator may yield different plausible topic labels for the same input x, enabling empirical estimation of the label distribution via Monte Carlo sampling.

4.1.2 Estimating $P^*(A)$ using Monte Carlo sampling

Please see **Algorithm 1** below:

Algorithm 1 Monte Carlo Estimation of $P^*(A)$

Require: Unlabeled target texts $\{x_i\}_{i=1}^{N}$, stochastic generator $G(a \mid x)$, number of samples S

Initialize counter $C(a) \leftarrow 0$ for all possible labels a
for $i = 1$ to N **do**
    for $s = 1$ to S **do**
        Sample $a \sim G(a \mid x_i)$
        Increment $C(a) \leftarrow C(a) + 1$
    end for
end for
Normalize: $P^*(a) \leftarrow \frac{C(a)}{N \cdot S}$ for all a
return Estimated distribution $P^*(A)$

4.2 Constructing $P(Y \mid A, X)$ and $P(Y \mid X)$

We use BERT (Bidirectional Encoder Representations from Transformers) for our classification task. BERT is a bidirectional transformer model pre-trained on large-scale unlabeled text by jointly conditioning on both left and right context in all layers [13]. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer for a wide range of tasks, including classifications.

For our experiments, we fine-tune the `bert-base-uncased` model from Hugging Face using the `BertForSequenceClassification` architecture for binary classification [14]. The input text X, along with the conditioning variable A (for modeling $P(Y \mid X, A)$ only), is combined, tokenized and passed through the pretrained BERT encoder, followed by a classification head to predict the label Y. Training is handled using Hugging Face's `Trainer` and `TrainingArguments` to estimate both $P(Y \mid X, A)$ and $P(Y \mid X)$ [15].

Putting it together, we now can now obtain $P^*(Y \mid do(x)) = P(Y \mid A, X)$ * $\mathrm{P}^*(A)$.

5 Experiment

Dataset. We use the UCB D-Lab: Measuring Hate Speech Dataset for our experiments. This dataset contains 39,565 labeled hate speech texts collected from Twitter, Reddit, and

YouTube [16].

We designate Twitter as the source domain for estimating $P(Y \mid X, A)$, and apply the procedure outlined in **Section 4** to estimate $P^*(Y \mid \text{do}(X))$ for data from Reddit and Youtube. We then compare this causal classifier against a baseline model trained on $P(Y \mid X)$. For the purposes of this experiment, and due to GPU limit, we define A as the topic of the text. In practice, A could be extended to include additional factors such as sentiment and style. Our results show a modest improvement in classification performance when using the causal approach. Please refer to `https://github.com/amynguyenpham/CI2_Project/tree/main` for data and training code.

Platform	$P(Y \mid X)$	$P^*(Y \mid \text{do}(X))$
Twitter	**99%**	95%
Reddit	79%	**80%**
Youtube	77%	**79%**

Table 2: Comparison of probabilistic classifier $P(Y \mid X)$ and transportable classifier $P^*(Y \mid \text{do}(X))$ across platforms

Table 2 compares the classification accuracy of the associational model $P(Y \mid X)$ and the transportable causal model $P^*(Y \mid \text{do}(X))$ across three platforms. While the transportable causal model's accuracy slightly decrease in domain (99% vs. 95%), the causal approach yields improved accuracy on Reddit (80% vs. 79%) and YouTube (79% vs. 77%), indicating better generalization in out-of-domain settings.

6 Conclusion

This paper examines hate speech classification through the lens of causal transportability, showing that models trained on a single platform often fail to generalize due to spurious, non-transferable correlations. Using causal graphs, we identify when the causal effect $P(Y \mid \text{do}(X))$ can be transferred across domains and used as a proxy for hate speech classifier. Our proposed method estimates this effect and yields modest gains in cross-platform accuracy. These results highlight the potential of causal inference to improve the generalization of text classifiers.

References

1. Gorwa, R., Binns, R., & Katzenbach, C. (2020). Algorithmic content moderation. *Big Data & Society*, 7(1). https://doi.org/10.1177/2053951719897945

2. Vanja Mladen Karan and Jan Šnajder. 2018. Cross-Domain Detection of Abusive Language Online. In Proceedings of the 2nd Workshop on Abusive Language Online (ALW2), pages 132–137, Brussels, Belgium. Association for Computational Linguistics.

3. C. Mao et al., "Causal Transportability for Visual Recognition," in 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA, 2022, pp. 7511-7521, doi: 10.1109/CVPR52688.2022.00737.

4. Reuters. (2023, May 11). Respect African content moderators, says Facebook whistleblower. Reuters. https://www.reuters.com/article/markets/commodities/respect-african-content-moderators-says-facebook- whistleblower-idUSL8N370520/

5. Tommaso Caselli, Valerio Basile, Jelena Mitrović, and Michael Granitzer. 2020. Hatebert: Retraining Bert for abusive language detection in english. arXiv preprint arXiv:2010.12472 (2020).

6. Judea Pearl. Causality: Models, reasoning, and inference, 2000.

7. Pearl, J., Bareinboim, E. (2011). Transportability of Causal and Statistical Relations: A Formal Approach (Technical Report R-372). UCLA Cognitive Systems Laboratory.

8. Bareinboim, E., & Pearl, J. (2013). Transportability of experimental results. *Journal of Causal Inference*, 1(1).

9. Ghenai, A., Noorian, Z., Moradisani, H., Abadeh, P., Erentzen, C., & Zarrinkalam, F. (2025). Exploring hate speech dynamics: The emotional, linguistic, and thematic impact on social media users. *Information Processing & Management*, 62(3), 104079. https://doi.org/10.1016/j.ipm.2025.104079

10. Rösner, L., & Krämer, N. C. (2016). Verbal venting in the social web: Effects of anonymity and group norms on aggressive language use in online comments. *Social Media + Society*, 2, 2056305116664220.

11. Sengupta, A., Bhattacharjee, S. K., Akhtar, M. S., & Chakraborty, T. (2022). Does aggression lead to hate? Detecting and reasoning offensive traits in Hinglish code-mixed texts. *Neurocomputing*, 488, 598–617.

12. Hugging Face. (2025). google/flan-t5-base [Model]. https://huggingface.co/google/flan-t5-base

13. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv:1810.04805.* https://doi.org/10.48550/arXiv.1810.04805

14. Hugging Face. (n.d.). *BERT model — Transformers documentation.* https://huggingface.co/docs/transformers/en/model_doc/bert

15. Hugging Face. (2020). *Trainer — Transformers 3.1.0 documentation.* https://huggingface.co/transformers/v3.1.0/training.html#trainer

16. Kennedy, C. J., Bacon, G., Sahn, A., & von Vacano, C. (2020). *Constructing interval variables via faceted Rasch measurement and multitask deep learning: A hate speech application. arXiv preprint arXiv:2009.10277.*